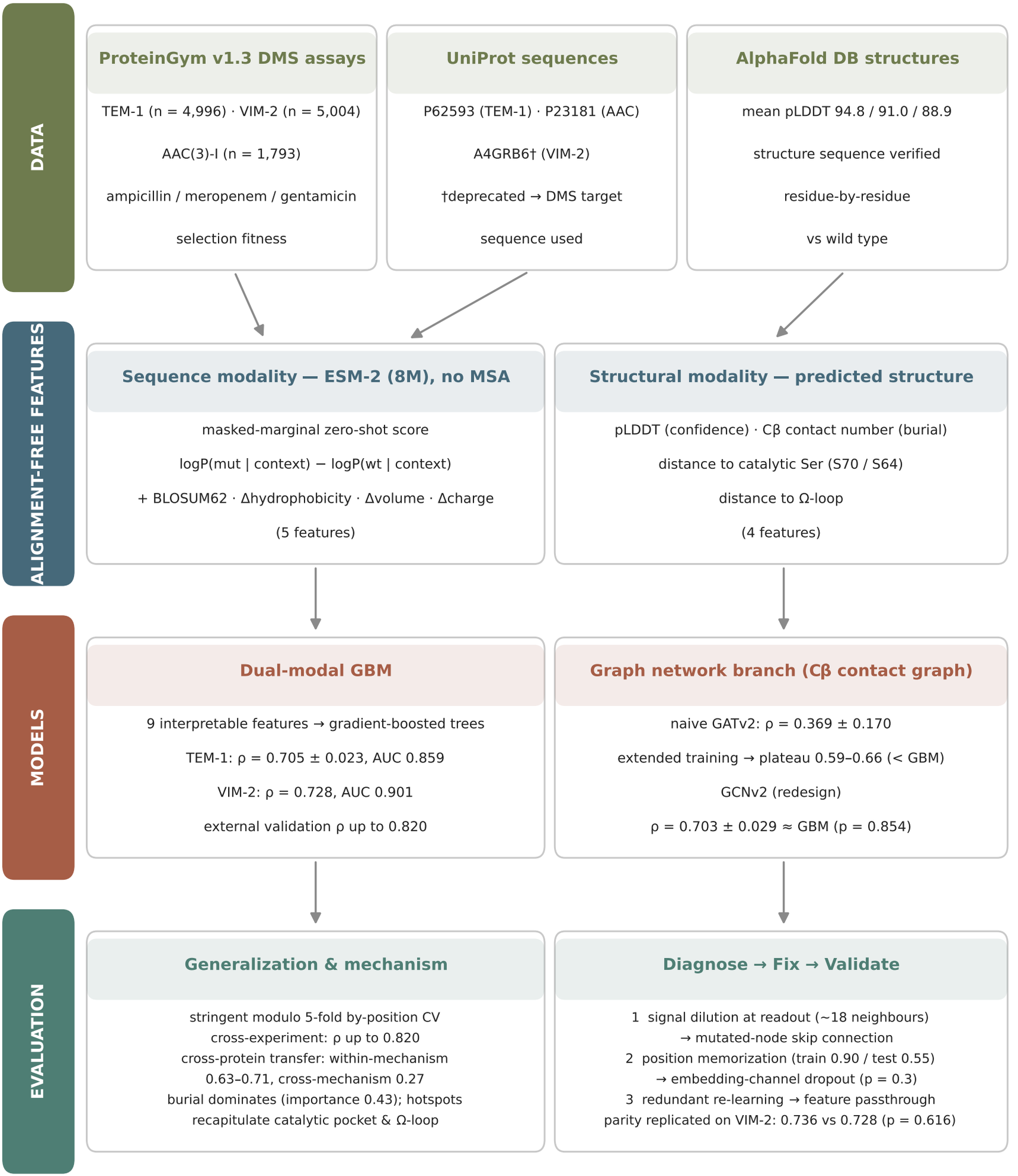
DATA
ProteinGym v1.3 DMS assays
TEM-1 (n = 4,996) · VIM-2 (n = 5,004)
AAC(3)-I (n = 1,793)
ampicillin / meropenem / gentamicin
selection fitness
UniProt sequences
P62593 (TEM-1) · P23181 (AAC)
A4GRB6† (VIM-2)
†deprecated → DMS target
sequence used
AlphaFold DB structures
mean pLDDT 94.8 / 91.0 / 88.9
structure sequence verified
residue-by-residue
vs wild type
ALIGNMENT-FREE FEATURES
Sequence modality — ESM-2 (8M), no MSA
masked-marginal zero-shot score
logP(mut | context) − logP(wt | context)
+ BLOSUM62 · Δhydrophobicity · Δvolume · Δcharge
(5 features)
Structural modality — predicted structure
pLDDT (confidence) · Cβ contact number (burial)
distance to catalytic Ser (S70 / S64)
distance to Ω-loop
(4 features)
MODELS
Dual-modal GBM
9 interpretable features → gradient-boosted trees
TEM-1: ρ = 0.705 ± 0.023, AUC 0.859
VIM-2: ρ = 0.728, AUC 0.901
external validation ρ up to 0.820
Graph network branch (Cβ contact graph)
naive GATv2: ρ = 0.369 ± 0.170
extended training → plateau 0.59–0.66 (< GBM)
GCNv2 (redesign)
ρ = 0.703 ± 0.029 ≈ GBM (p = 0.854)
EVALUATION
Generalization & mechanism
stringent modulo 5-fold by-position CV
cross-experiment: ρ up to 0.820
cross-protein transfer: within-mechanism
0.63–0.71, cross-mechanism 0.27
burial dominates (importance 0.43); hotspots
recapitulate catalytic pocket & Ω-loop
Diagnose → Fix → Validate
1 signal dilution at readout (~18 neighbours)
→ mutated-node skip connection
2 position memorization (train 0.90 / test 0.55)
→ embedding-channel dropout (p = 0.3)
3 redundant re-learning → feature passthrough
parity replicated on VIM-2: 0.736 vs 0.728 (p = 0.616)